

Human Behavior FCA ■ Functional Causal Architecture for Human Behavior Generation

1. Core Definition

Human Behavior FCA is a Functional Causal Architecture for Human Behavior Generation. It explains how behavior is generated, stabilized, interrupted, locked in, and updated through reality-facing feedback.

2. Minimal Behavior Generation Chain

Observable Output

→ Foreground Entry

→ Historical Template Matching and Explanation

→ New Structure Generation

→ Execution Maintenance

→ Feedback Update

3. Four Functional Systems

3.1 SN ■ Salience / Foreground Entry System

Determines which inputs enter foreground processing.

3.2 DMN ■ Historical Template Matching and Explanation System

Maps current input onto stored templates to maintain continuity of world, self, identity, relationship, and context. DMN does not generate genuinely new structures.

3.3 FP ■ New Structure Generation Layer

Generates new causal paths, new action structures, and new solution spaces beyond stored templates. FP is a functional generation layer, not an isolated anatomical brain network.

3.4 ECN ■ Execution Maintenance System

Maintains generated paths as long-term action, repetition load, and reality-facing output.

4. Hormone Modulation Layer

Hormones are modulation axes, not networks, and not one-to-one hard correspondences.

- Cortisol: raises the probability that threat-related input enters the foreground and increases DMN threat-template weight.

- Dopamine: raises FP occupancy stability and supports exploration, path search, and new-structure generation.
- Oxytocin: lowers social threat sensitivity and supports continuity, trust background, and ECN maintenance stability.

5. Breakdown and Lock-in

Behavior breakdown is not a personality problem or a motivation problem. FCA localizes the breakpoint through SN, DMN, FP, ECN, and feedback update.

6. Application Scope

- Relationship dynamics
- Decision formation
- Execution stability
- Behavioral repetition
- AI Runtime personalization

7. Boundary

FCA is not a personality typology, not a psychological labeling system, not a motivation theory, not a moral judgment framework, and not a chatbot wrapper.

8. Public Layer Boundary

The public layer fixes definitions, canonical terms, chains, and boundaries. Internal runtime may include samples, scoring, parameters, and admin models, but those do not enter the public definition source.

9. Canonical Source

Canonical web definition source:

<https://alitaos.com/en/human-behavior-fca>

Versioned source archive:

<https://github.com/dujf921/afa-triplex-modulation-architecture>